

Project Description and Methodology Overview

1. Problem Formulation

The objective of this study is to develop a robust and adaptive regression framework for predicting:

- **motor_UPDRS**
- **total_UPDRS**

from biomedical voice measurements.

Let:

- $X \in \mathbb{R}^{n \times d}$ denote the feature matrix
- $y \in \mathbb{R}^{n \times 2}$ denote the two-target regression outputs

The problem is formulated as a **multi-output nonlinear regression task**, where the goal is to learn a function:

$$f : \mathbb{R}^d \rightarrow \mathbb{R}^2$$

that minimizes predictive error under mean squared loss.

However, due to heterogeneity in voice-derived biomarkers and varying nonlinear relationships across disease severity, a single global model may not be sufficient. Therefore, an adaptive ensemble mechanism is proposed.

2. Data Processing Pipeline

2.1 Feature Selection and Cleaning

- Non-informative identifiers were removed.
- Biomedical voice features were retained.
- No target leakage variables were included.

2.2 Dataset Splitting

The dataset was split into:

- Training set
- Validation set
- Test set

Splits were frozen and exported to ensure full reproducibility and to prevent re-splitting variance across experiments.

2.3 Feature Scaling

A `StandardScaler` was applied to features:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

Scaling was fit only on training data and reused for validation and test data.

2.4 Target Scaling

Targets were scaled internally during training for neural models to improve optimization stability. Final evaluations were performed on inverse-transformed predictions to ensure clinical interpretability.

3. Expert Model Architecture

Four heterogeneous expert models were independently trained:

3.1 Neural Network Expert

- Fully connected feedforward network
- Multi-layer architecture with ReLU activations
- Dropout regularization
- L2 weight decay
- Optimized using Adam
- Hyperparameters tuned via Optuna

This expert captures nonlinear feature interactions.

3.2 Random Forest Expert

- Ensemble of decision trees
- Bootstrapped aggregation
- Hyperparameters optimized via Optuna

This expert captures nonlinear, piecewise-constant relationships and feature threshold effects.

3.3 Ridge Regression Expert

- Linear regression with L2 regularization
- Multi-output formulation

This expert provides a bias-controlled linear baseline.

3.4 Elastic Net Expert

- Multi-task Elastic Net
- Combined L1 and L2 regularization
- Encourages sparse yet stable solutions

This expert balances sparsity and multicollinearity handling.

Each expert was trained independently and then frozen. No joint training was performed between experts and the gating mechanism.

4. Mixture of Experts Framework

4.1 Motivation

Instead of static averaging, a dynamic weighting mechanism was introduced to adaptively combine experts per input sample.

The core idea:

Different regions of feature space may favor different predictive mechanisms.

4.2 Meta-Feature Construction

For each datapoint:

$$Z = [E_1(x), E_2(x), E_3(x), E_4(x)]$$

Where each ($E_i(x)$) produces a 2-dimensional prediction.

Thus:

$$Z \in \mathbb{R}^{n \times 8}$$

This forms the input to the gating network.

4.3 Gating Network Architecture

A lightweight neural network was trained to learn adaptive weights:

- Input: concatenated expert predictions
- Output: 8 weights
 - 4 weights for motor_UPDRS
 - 4 weights for total_UPDRS

Weights were passed through Softmax to enforce convex combinations:

$$\sum_i w_i(x) = 1$$

Separate weight distributions were learned for each target.

4.4 Final Prediction Rule

For each target:

$$\hat{y}_{motor} = \sum_{i=1}^4 w_{motor,i}(x) \cdot E_{i,motor}(x)$$

$$\hat{y}_{total} = \sum_{i=1}^4 w_{total,i}(x) \cdot E_{i,total}(x)$$

This ensures:

- Interpretability
 - Stability
 - Target-specific specialization
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4.5 Entropy Regularization

To prevent expert collapse (i.e., domination by a single expert), entropy regularization was introduced:

$$L_{total} = L_{MSE} - \lambda \cdot H(w)$$

Where:

$$H(w) = - \sum_i w_i \log w_i$$

This encourages diversity in expert contributions while preserving predictive accuracy.

5. Optimization Strategy

- Loss: Mean Squared Error
- Early stopping based on validation MSE
- Adam optimizer
- Hyperparameter tuning via Optuna
- GPU acceleration for neural components

No information from the test set was used during training or model selection.

6. Evaluation Metrics

Performance was evaluated using:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

Residual distributions and weight distributions were analyzed to assess:

- Bias
 - Variance
 - Expert specialization behavior
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7. Interpretability Analysis

The gating mechanism was analyzed by computing:

- Mean expert weight per target
- Weight variance
- Distribution of expert dominance

This provided insight into:

- Which experts specialize in which targets
 - How nonlinear and linear models contribute differently
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8. Key Observations

- Random Forest dominated in most feature regions.
 - Linear models contributed modest but consistent corrections.
 - Neural network contribution varied depending on entropy strength.
 - Adaptive weighting improved generalization over static ensembling.
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9. Reproducibility Measures

To ensure strict reproducibility:

- Frozen dataset splits
 - Saved scalers
 - Saved expert parameters
 - Exported intermediate meta-datasets
 - Version-locked dependencies
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10. System Design Philosophy

The architecture was designed to balance:

- Predictive performance
- Interpretability
- Stability
- Scientific defensibility

Rather than relying on a single highly complex model, the framework leverages complementary modeling assumptions through adaptive coordination.